EDA & Feature Engineering

What We Did and What We Found

M2 Secondary Meta-Labeling | Branch: sam_M2

Raviv - Samweg - Cristian | Week of Mar 16, 2026

FEATURE ENGINEERING

Turning raw VIX and OAS data into 10 meaningful inputs the M2 model can learn from.

EDA

Testing whether those features actually relate to M1 winning before building any model.

Key finding: neither VIX nor OAS significantly predicts M1 performance

PART 1: FEATURE ENGINEERING

How we turned raw data into model inputs

What is Feature Engineering?

The problem with raw data

A model cannot learn from raw numbers like 'VIX = 28.3' alone. It has no context - is 28.3 high or low? Is it getting worse or better? Has it been like this for months, or did it just spike?

Feature engineering solves this

We transform raw VIX and OAS data into 5 meaningful features each - ones that carry context the model can actually use.

The no-lookahead rule

Every single feature is computed using only past data. The z-score at March 2020 uses only data from before March 2020. This ensures the model never cheats by seeing future information during training.

RESULT: 10 features engineered from 2 raw data sources (VIX + OAS). Zero lookahead bias.

VIX Features - Measuring Market Fear

What is VIX?

VIX is the 'fear index' - it measures how much volatility investors expect in the stock market. High VIX = panic and uncertainty. Low VIX = calm markets.

The 5 features we built from VIX:

vix_level_z How high is VIX vs its own history? (+1.8 = 1.8 std devs above normal)

vix_change_z How much did VIX move this month, vs typical monthly moves?

vix_trend Is VIX above or below its 6-month rolling average?

vix_high_regime Is VIX above its all-time expanding median? (1 = yes, 0 = no)

vix_rising Did VIX go up this month? (1 = yes, 0 = no)

Why these specific features?

Together they answer: where is fear (level), how fast is it changing (change), which direction is it trending (trend), and are we in a sustained fear regime vs a one-off spike (regime flag).

RESULT: 5 VIX features capture: level, momentum, trend, regime, and direction of market fear.

OAS Features - Measuring Credit Stress

What is OAS?

OAS (Option-Adjusted Spread) measures the extra yield investors demand to hold corporate bonds instead of safe government bonds. Wide OAS = credit stress and illiquidity. Tight OAS = easy credit conditions.

The 5 features we built from OAS:

oas_level_z	How wide are spreads vs their own history?
oas_change_z	How much did spreads move this month?
oas_trend	Are spreads above or below their 6-month average?
oas_wide_regime	Are spreads in a historically wide (stressed) regime?
oas_widening	Did spreads widen this month? (credit getting worse?)

The hypothesis for using OAS

When credit spreads are wide and widening, it signals financial stress that might cause M1's signals to be less reliable - similar logic to VIX.

RESULT: 5 OAS features capture: level, momentum, trend, regime, and direction of credit stress.

PART 2: EDA

Testing whether the features actually matter

What is EDA and Why Did We Do It?

EDA = Exploratory Data Analysis

Before building any model, we asked a simple question: does VIX or OAS actually relate to whether M1's signals win or lose? If they don't, adding them to M2 will only add noise.

The hypothesis we were testing

M1 follows market trends. In calm markets (low VIX, tight OAS), trends are stable and M1 should win more often. In fearful, stressed markets (high VIX, wide OAS), trends break down and M1 should win less often.

What we actually measured

We split 138 M1 events into two groups - high and low VIX - and compared M1's win rate in each group. We then ran statistical tests to check if any difference was real or just random chance.

The standard for 'real'

A p-value below 0.05 means there is less than 5% chance the difference is random. A p-value above 0.05 means we cannot rule out that it is just noise.

RESULT: 6 EDA charts were produced. All pointed to the same conclusion.

Chart 1 - Does VIX Predict M1 Wins?

What this chart shows

M1's win rate split by Low VIX (calm markets) vs High VIX (fearful markets), across all 138 events. Also broken out by BUY and SELL signals separately.

The numbers

Low VIX: 67.9% win rate (n=53 events)

High VIX: 70.6% win rate (n=85 events)

The difference is only 2.7 percentage points - and it goes the wrong direction. M1 actually wins slightly MORE when VIX is high.

The statistical test

Chi-square p-value = 0.8882. This means there is an 89% chance this small difference is pure random noise. Far above the 0.05 threshold needed to call it meaningful.

RESULT: VIX regime has no statistically significant effect on M1's win rate (p=0.8882).

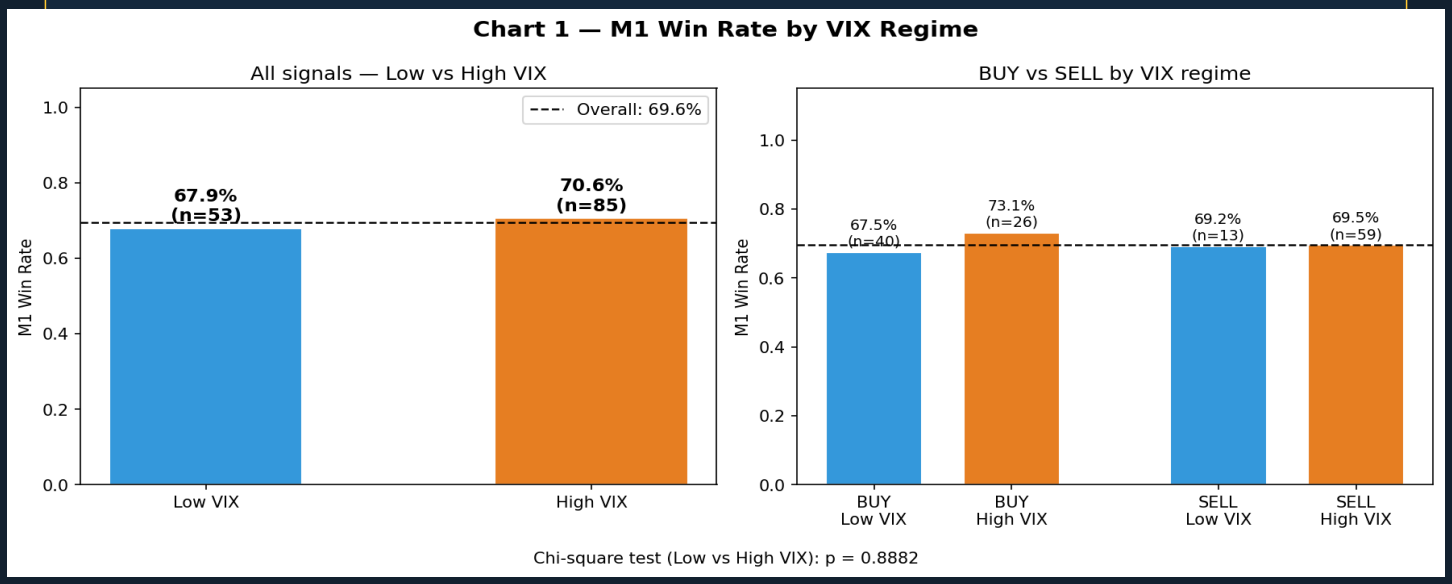

Chart 1: M1 Win Rate by VIX Regime. Chi-square p=0.8882 - no significant relationship.

Chart 2 - Does OAS Predict M1 Wins?

What this chart shows

Same test but for OAS. M1's win rate in Tight OAS (easy credit) vs Wide OAS (credit stress) markets.

The numbers

Tight OAS: 66.7% win rate (n=42 events)

Wide OAS: 70.8% win rate (n=96 events)

Again, the direction is backwards - M1 wins more when credit is stressed. The hypothesis said stressed markets should hurt M1, but the data disagrees.

The statistical test

Chi-square p-value = 0.7730. 77% chance this is random noise. Not significant.

RESULT: OAS regime has no statistically significant effect on M1's win rate (p=0.7730).

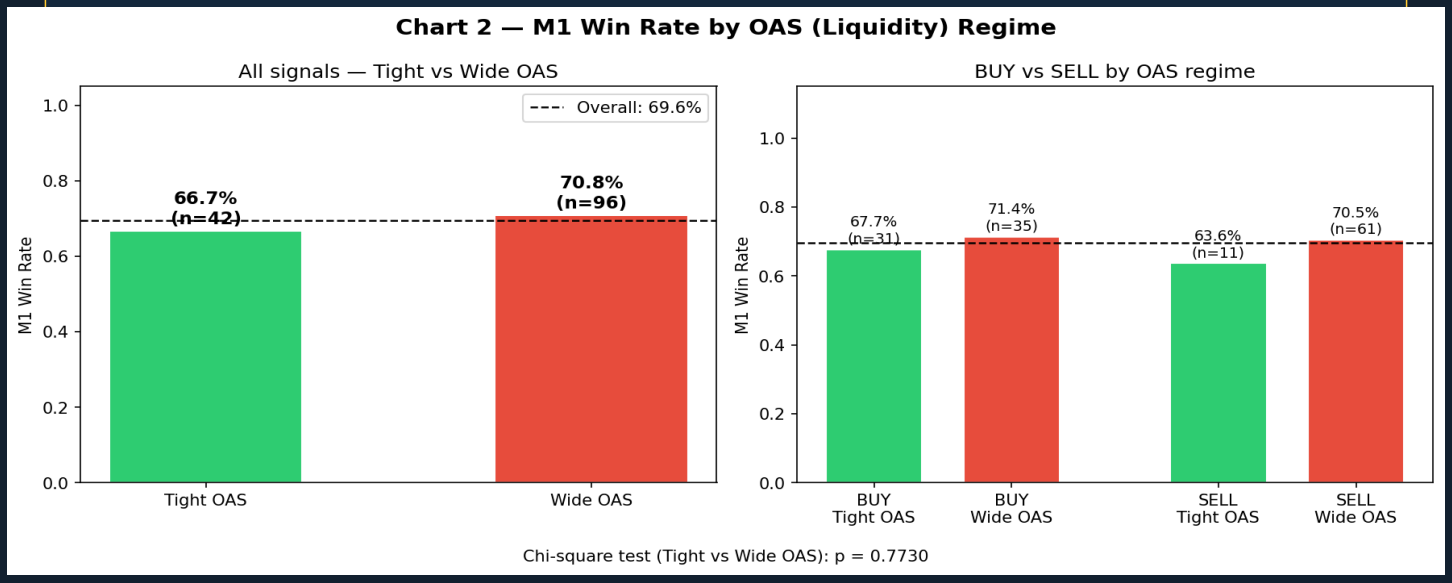

Chart 2: M1 Win Rate by OAS Regime. Chi-square p=0.7730 - no significant relationship.

Chart 3 - VIX and OAS Combined

What this chart shows

A heatmap of M1's win rate and average return across all four combinations of VIX and OAS regimes simultaneously.

The most striking finding

The supposedly 'worst' regime - High VIX + Wide OAS (fearful market, stressed credit) - actually has the BEST performance: 72.7% win rate and +1.02% avg monthly return.

The supposedly 'best' regime - Low VIX + Tight OAS (calm, easy credit) - had only 70.6% win rate and just +0.31% avg monthly return.

What this means

M1 is not just robust to market stress - it may actually perform better during stress, possibly because its SELL signals correctly call downturns. This makes VIX and OAS useless as filters.

RESULT: High VIX + Wide OAS (worst expected regime) delivered the best actual M1 performance.

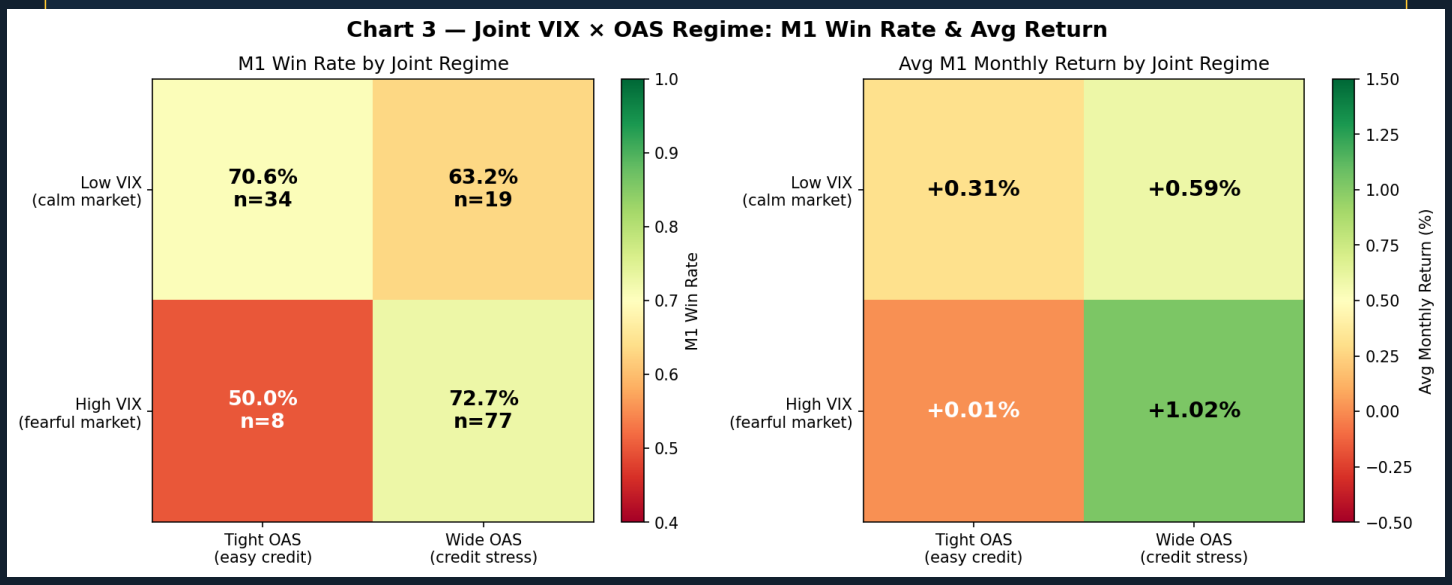

Chart 3: Joint VIX x OAS heatmap. Stressed regimes show equal or better M1 performance.

Chart 4 - Feature Correlations with M1 Wins

What this chart shows

The correlation of every single feature with the meta_label (did M1 win?). A perfect predictor would show +1.0 or -1.0. A useless feature shows 0.

The key finding

The highest correlation is only +0.108 (trailing average return) - essentially zero predictive power. All VIX and OAS features sit between -0.03 and +0.07.

Importantly, none of the correlations have a * or ** significance marker, meaning none are statistically significant at p<0.05.

What this confirms

No individual feature - including VIX and OAS - has a meaningful linear relationship with M1 winning. The signal simply is not there in the data.

RESULT: Highest correlation with M1 wins: 0.108. All VIX/OAS features below 0.07.

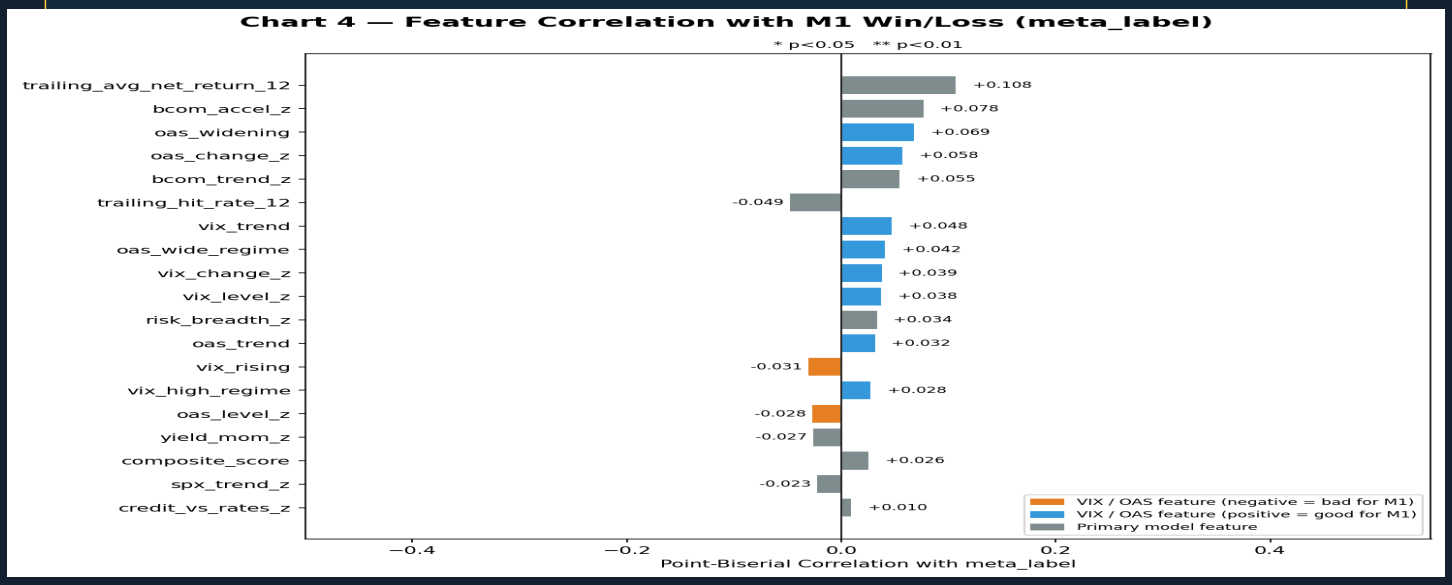

Chart 4: All feature correlations with meta_label are below 0.11 - none statistically significant.

Charts 5 & 6 - Return Distributions and Regime Gate

Chart 5: Return distributions by regime (t-test)

Compared the full distribution of M1 monthly returns in High vs Low VIX. High VIX averaged +0.93%, Low VIX averaged +0.41% - but the t-test p-value was 0.1597. Not significant. The distributions overlap too much.

Chart 6: What if we used VIX/OAS as a simple gate?

Simulated trading only in 'favorable' regimes (Low VIX + Tight OAS) vs trading only in 'stressful' regimes (High VIX + Wide OAS):

All events (no filter): Sharpe 1.19, Win Rate 69.6%

Low VIX only: Sharpe 0.84, Win Rate 67.9% (worse)

Low VIX + Tight OAS: Sharpe 0.68, Win Rate 70.6% (much worse)

High VIX + Wide OAS: Sharpe 1.49, Win Rate 72.7% (best!)

Filtering out stressed markets actually destroyed value. The 'bad' regime was the best-performing one.

RESULT: Simple regime gate using VIX+OAS reduces Sharpe from 1.19 to 0.68. No improvement found.

Chart 6 — Simulated M2 Regime Gate: If You Only Traded in Favorable Regimes

Performance Under Different Regime Filters

Filter	N events	Win Rate	Avg Return	Sharpe	Max Loss
All M1 events (no filter)	138	69.6%	+0.73%	1.19	-5.39%
Low VIX only	53	67.9%	+0.41%	0.84	-3.32%
Tight OAS only	42	66.7%	+0.25%	0.55	-3.32%
Low VIX + Tight OA (best regime)	34	70.6%	+0.31%	0.68	-3.32%
High VIX + Wide OA (worst regime)	77	72.7%	+1.02%	1.49	-5.39%

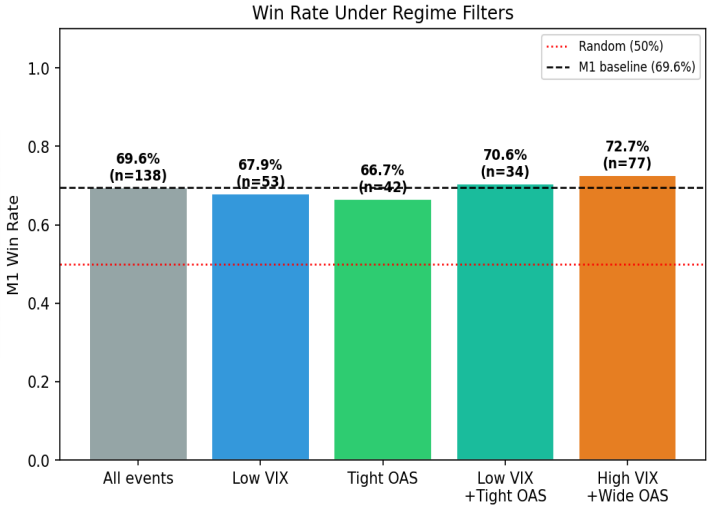

Chart 6: Regime gate simulation. Filtering by 'favorable' VIX/OAS conditions hurt performance.

Summary & Conclusions

FEATURE ENGINEERING

We took raw monthly VIX and OAS data and engineered 10 features:

VIX: level z-score, change z-score, 6-month trend, high-regime flag, rising flag

OAS: level z-score, change z-score, 6-month trend, wide-regime flag, widening flag

All z-scores used only past data - no future information leaked into any feature.

EDA FINDINGS

Chi-square test (VIX vs M1 win rate): $p = 0.8882$ - no significant relationship

Chi-square test (OAS vs M1 win rate): $p = 0.7730$ - no significant relationship

T-test on returns by VIX regime: $p = 0.1597$ - not significant

T-test on returns by OAS regime: $p = 0.0787$ - not significant

All feature correlations with meta_label were below 0.11 - essentially zero.

M2 MODEL RESULTS

Logistic regression trained walk-forward on 60-month expanding windows.

OOS accuracy: 53.9% | ROC-AUC: 0.523 (random = 0.50)

M1 baseline Sharpe: 1.31 -> M2-filtered Sharpe: 1.18 (got worse)

M2-rejected trades had higher win rate than M2-approved trades.

CONCLUSION

VIX and OAS features are theoretically sound but carry no detectable signal

for predicting M1 wins over 28 years of monthly data.

Most likely explanation: M1's own indicators (SPX trend, credit vs rates,

risk breadth) already absorb the stress information that VIX captures.

Next step: find features that are orthogonal to M1's existing inputs.